

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			3 = 2 marks; 2 = 1 mark: I E II A III C		2		2		2
	(b)			(apoplast) water travels through the cell walls (1) (symplast) water travels through cytoplasm <u>and</u> plasmodesmata (1)	2			2		
	(c)	(i)		Any four (×1) from A. {Casparian strip / suberin} {blocks the apoplast pathway / forces water into the symplast pathway} (1) B. Active transport of ions into the {endodermis/ xylem} (1) C. This lowers the water potential in the {endodermis/ xylem} (1) D. Causing water to move into the {endodermis/ xylem} by <u>osmosis</u> (1) E. Producing <u>hydrostatic</u> pressure (1)	2	2		4		
		(ii)		Any three (×1) from A. Cyanide prevents {(aerobic) respiration / ATP production} / Cyanide is a respiratory inhibitor (1) B. (No ATP) so <u>no</u> active transport of {ions/ solutes} (1) C. Water potential of the {endodermis/ xylem} is not lowered / water potential gradient not established (1) D. So, water not taken up by xylem (1) OR if referring to root hair cell C. Water potential of the root hair cell is not lowered / water potential gradient not established (1) D. So, water not taken up by the {roots/root hair cells} (1)	2		1	3		3
				Question 6 total	6	4	1	11	0	5